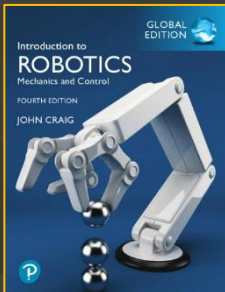
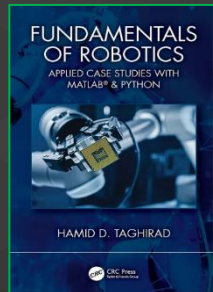


ARAS Robotics & Computer Vision



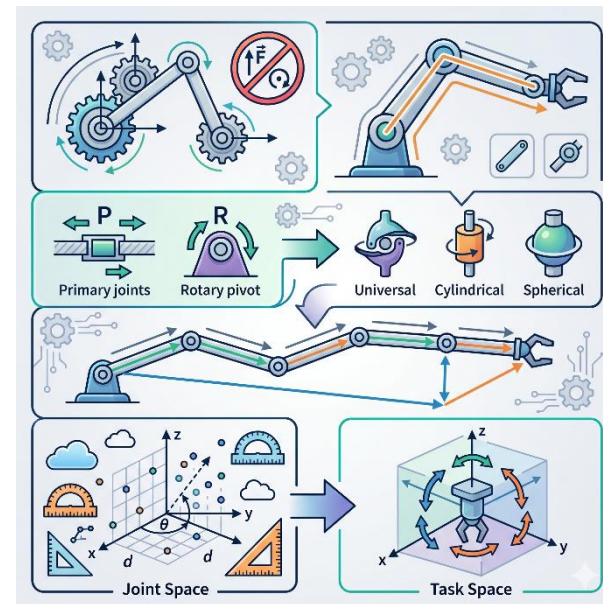
Chapter 3: Kinematics fundamentals

joint space vs. task space; simplified DH convention; computing end-effector pose; high-level overview of inverse kinematics, dynamics, and control.



Definitions

- ✓ The study of the **geometry** of motion in a robot, **without** considering the forces and torques that **cause** the motion.
- ✓ A serial robot consist of
 - A single kinematic loop
 - A number of links and joints
 - The joints might be primary (P or R) or compound (U, C, S)
- ✓ **Kinematic loop closure**
 - A loop consists of the consecutive links and joint to the end-effector
 - **Rigid links with primary joints**
 - Compound joints are reduced to a number of primary joints
 - The loop is written in a vector form
 - The joint motion variables form the **Joint Space**
 - The end-effector final motion DoF's form the **Task Space**



✓ Example: Elbow Manipulator

- 3DoF spatial manipulator (RRR)

Joint variables: $\mathbf{q} = [\theta_1 \quad \theta_2 \quad \theta_3]^T$

$$q_i = \begin{cases} \theta_i & \text{if joint } i \text{ is revolute} \\ d_i & \text{if joint } i \text{ is prismatic} \end{cases}$$

Task variables: $\mathbf{x} = [x_e \quad y_e \quad \theta_e]^T$

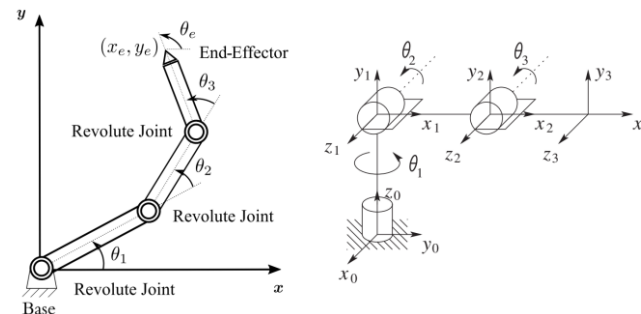
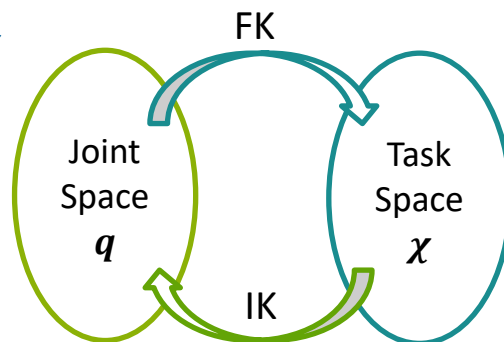
The position and orientation variables of end-effector

- Forward kinematics

Given $\mathbf{q}$ find $\mathbf{x}$

Inverse kinematics

Given $\mathbf{x}$ find $\mathbf{q}$



- Motivating Example

- ✓ Kinematic loop closure

- Assign base coordinate frame $\{0\}$

Denote $\mathbf{q} = [\theta_1, \theta_2]^T$ and $\mathbf{x} = [x_e, y_e]^T$

Denote the link vectors 

and the end-effector vector 

Write the loop closure vector equation:

$$\vec{a}_1 + \vec{a}_2 = \vec{e}$$

$$a_1 \cos \theta_1 + a_2 \cos(\theta_1 + \theta_2) = x_e$$

$$a_1 \sin \theta_1 + a_2 \sin(\theta_1 + \theta_2) = y_e$$

Shorthand notation (FK)

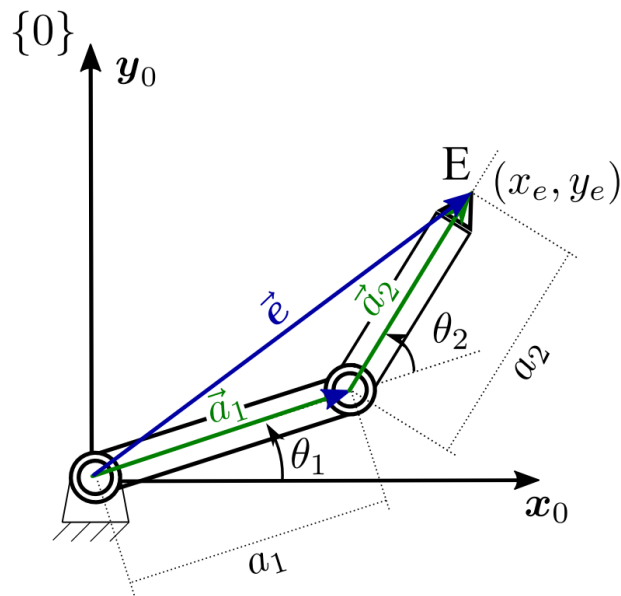
$$a_1 c_1 + a_2 c_{12} = x_e$$

$$a_1 s_1 + a_2 s_{12} = y_e$$

In which $c_1 = \cos \theta_1$, $s_1 = \sin \theta_1$, $c_{12} = \cos(\theta_1 + \theta_2)$, $s_{12} = \sin(\theta_1 + \theta_2)$.

Given joint variables $\mathbf{q} = [\theta_1, \theta_2]^T$, the task space variables $\mathbf{x} = [x_e, y_e]^T$ is found from FK formulation.

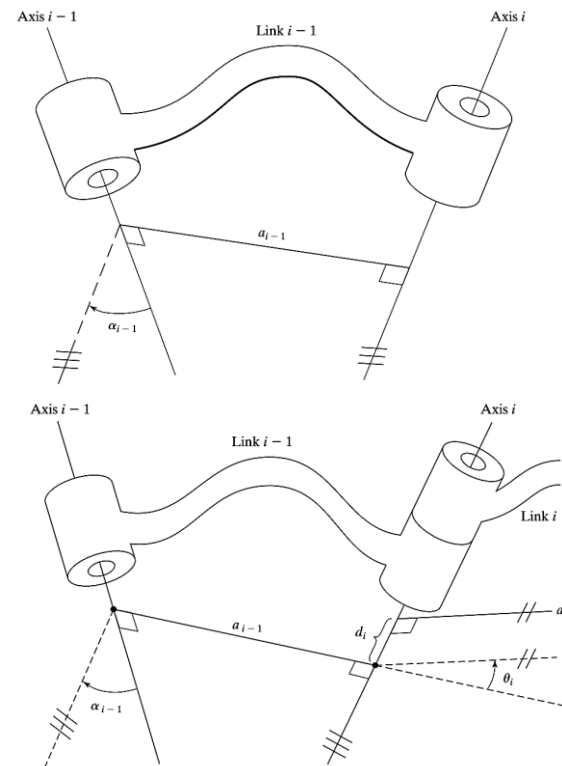
Inverse problem (IK) may be found by algebraic calculations.



- Algorithmic Approach

- ✓ General Link Parameters

- Link length a_i and link twist α_i
Start from zero frame based on the joint axes
Link length a_{i-1} common normal line lengths
Link twist α_{i-1} relative angle of two joint axes
 - Link offset d_i and joint angle θ_i
Neighboring link distance d_i and angle θ_i
For rotary joints $q_i = \theta_i$ is the joint variable
For prismatic joints $q_i = d_i$ is the joint variable
 - First and Last link in the chain
Consider base of the robot the 0 link and frame
For the last frame on the end-effector coplanar to the previous frame



- ✓ Frame Assignment (Paul's Convention)

- The $\hat{Z}_{i-1}$ axis of rotation of $\{i\}$ joint (R)
- OR the axis of translation of $\{i\}$ joint (P)
- The origin of frame $\{i\}$ is at the intersection of perpendicular line to the axis i .
- The $\hat{X}_i$ axis points along a_i along the common normal. In case a_i is zero $\hat{X}_i$ is normal to the plane of $\hat{Z}_{i-1}$ and $\hat{Z}_i$

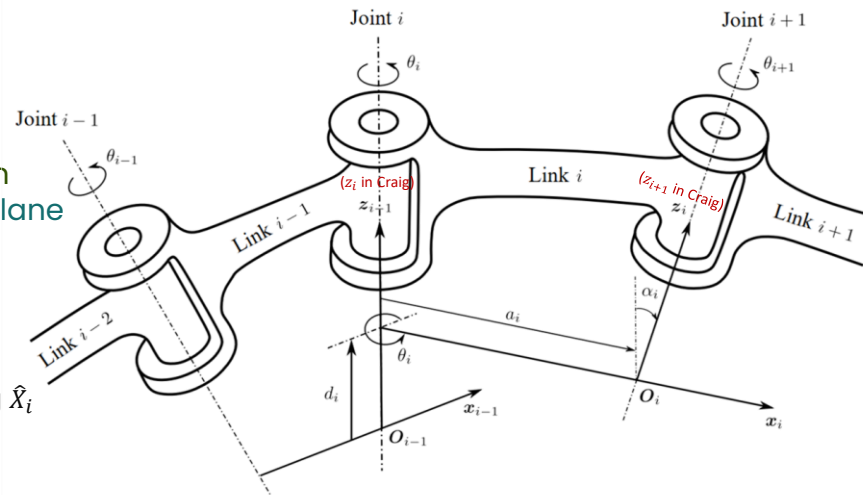
- ✓ Denavit-Hartenberg (DH) Parameters

a_i = The distance from $\hat{Z}_{i-1}$ to $\hat{Z}_i$ measured along $\hat{X}_i$

α_i = The angle from $\hat{Z}_{i-1}$ to $\hat{Z}_i$ measured about $\hat{X}_i$

d_i = The distance from $\hat{X}_{i-1}$ to $\hat{X}_i$ measured along $\hat{Z}_{i-1}$

θ_i = The angle from $\hat{X}_{i-1}$ to $\hat{X}_i$ measured about $\hat{Z}_{i-1}$.



Forward Kinematics

- Algorithmic Approach
 - ✓ Denavit-Hartenberg (DH) Parameters



- DH Homogeneous Transformations (Paul's Convention)

- Consider three intermediate Frames

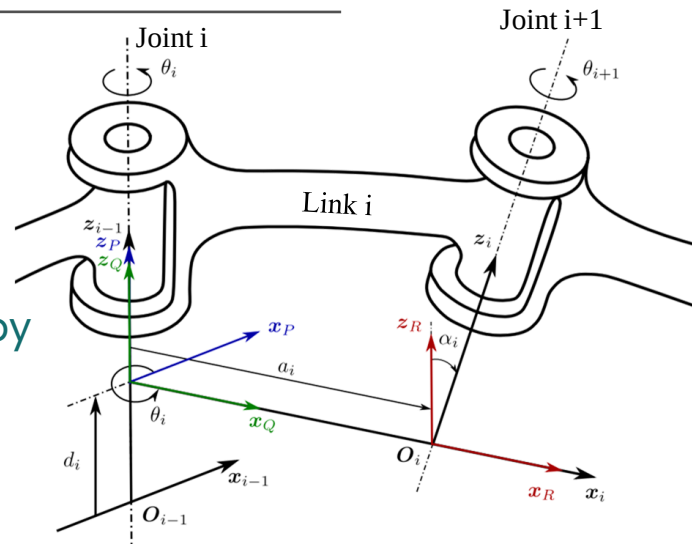
$\{i-1\}, \{P\}, \{Q\}, \{R\}, \{i\}$

The general transformation will be found by considering four transformation about moving frames (pre-multiplication)

$${}^{i-1}T = {}^{i-1}P {}^PT {}^PQ {}^PQ {}^QR {}^RT {}^RT {}^RT$$

In which:

$${}^{i-1}T = D_z(d_i)R_z(\theta_i)D_x(a_i)R_x(\alpha_i)$$



$${}^{i-1}T_P = T(z, d_i) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^PT_Q = T(z, \theta_i) = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & 0 \\ s\theta_i & c\theta_i & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

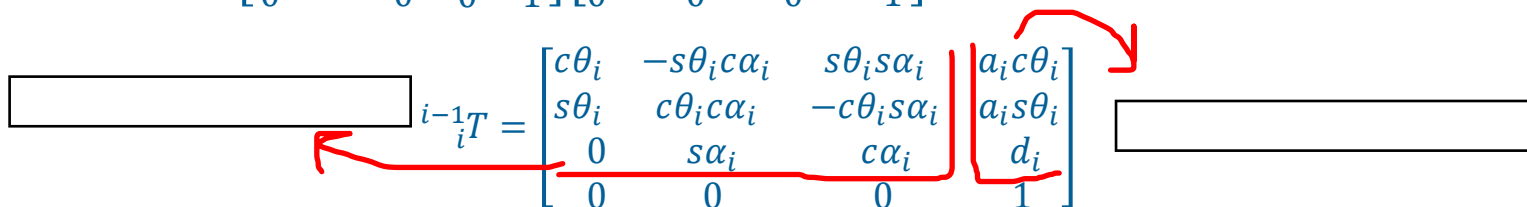
$${}^QT_R = T(x, a_i) = \begin{bmatrix} 1 & 0 & 0 & a_i \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^RT_i = T(x, \alpha_i) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & c\alpha_i & -s\alpha_i & 0 \\ 0 & s\alpha_i & c\alpha_i & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Forward Kinematics

- Algorithmic Approach
 - ✓ DH Homogeneous Transformations (Paul's Convention)

$${}^{i-1}_iT = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & 0 \\ s\theta_i & c\theta_i & 0 & 0 \\ 0 & 0 & 1 & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & a_i \\ 0 & c\alpha_i & -s\alpha_i & 0 \\ 0 & s\alpha_i & c\alpha_i & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



$${}^{i-1}_iT = \begin{bmatrix} c\theta_i & -s\theta_i c\alpha_i & s\theta_i s\alpha_i & a_i c\theta_i \\ s\theta_i & c\theta_i c\alpha_i & -c\theta_i s\alpha_i & a_i s\theta_i \\ 0 & s\alpha_i & c\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

And the inverse is

$${}^{i-1}_iT = \begin{bmatrix} c\theta_i & s\theta_i & 0 & -a_i \\ -s\theta_i c\alpha_i & c\theta_i c\alpha_i & s\alpha_i & -d_i s\alpha_i \\ s\theta_i s\alpha_i & -c\theta_i s\alpha_i & c\alpha_i & -d_i c\alpha_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Examples:

✓ Example 1: Planar RRR Manipulator

- Algorithmic Approach (Paul's Convention)

Denote $\mathbf{q} = [\theta_1, \theta_2, \theta_3]^T$ and $\mathbf{x} = [x_E, y_E, \theta_E]^T$

Affix the frames and find DH-parameters.

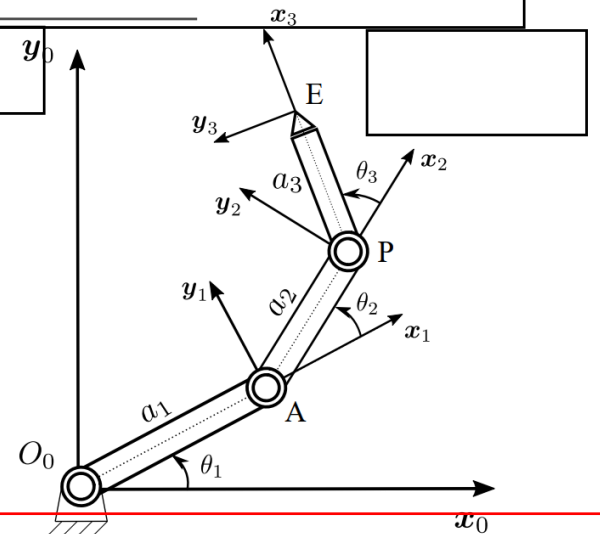
Find the homogeneous transformations:

$${}^0_1T = \begin{bmatrix} c_1 & -s_1 & 0 & a_1c_1 \\ s_1 & c_1 & 0 & a_1s_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad {}^1_2T = \begin{bmatrix} c_2 & -s_2 & 0 & a_2c_2 \\ s_2 & c_2 & 0 & a_2s_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

$${}^2_3T = \begin{bmatrix} c_3 & -s_3 & 0 & a_3c_3 \\ s_3 & c_3 & 0 & a_3s_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Find the loop closure equation in matrix form:

$$\underline{{}^0ET = {}^0_3T = {}^0_1T {}^1_2T {}^2_3T}$$



i	a_i	α_i	d_i	θ_i
1	a_1	0	0	θ_1
2	a_2	0	0	θ_2
3	a_3	0	0	θ_3

- Examples:

- ✓ Example 1: (Cont.)

- Algorithmic Approach (Paul's Convention)

Calculate the loop closure matrix equation:

$${}^0T_E = {}^0T_3 = {}^0T_1 {}^1T_2 {}^2T_3$$

$${}^0T_E = \begin{bmatrix} c\theta_E & -s\theta_E & 0 & x_E \\ s\theta_E & c\theta_E & 0 & y_E \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

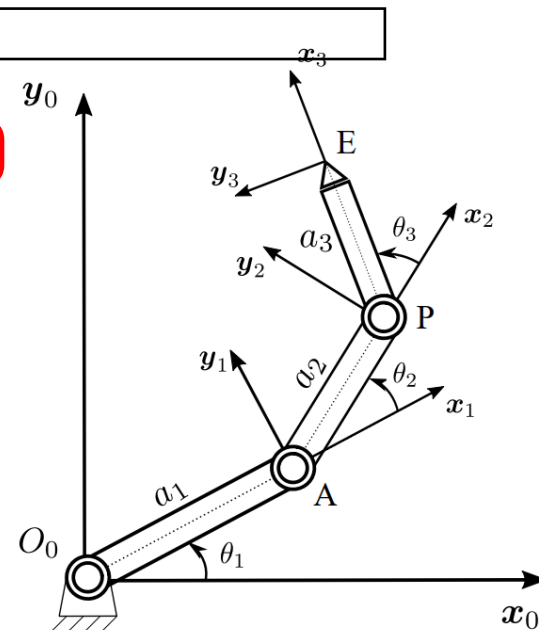
$${}^0T_3 = {}^0T_1 {}^1T_2 {}^2T_3 = \begin{bmatrix} c_{123} & -s_{123} & 0 & a_1c_1 + a_2c_{12} + a_3c_{123} \\ s_{123} & c_{123} & 0 & a_1s_1 + a_2s_{12} + a_3s_{123} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Shorthand notation (FK)

$$a_1c_1 + a_2c_{12} + a_3c_{123} = x_E$$

$$a_1s_1 + a_2s_{12} + a_3s_{123} = y_E$$

$$\theta_E = \theta_1 + \theta_2 + \theta_3$$



- Examples:
 - ✓ Example 2: 3DOF Elbow Manipulator

تکلیف تنظیمات تحویل کار نمره‌دهی پیشرفته بیشتر ▼

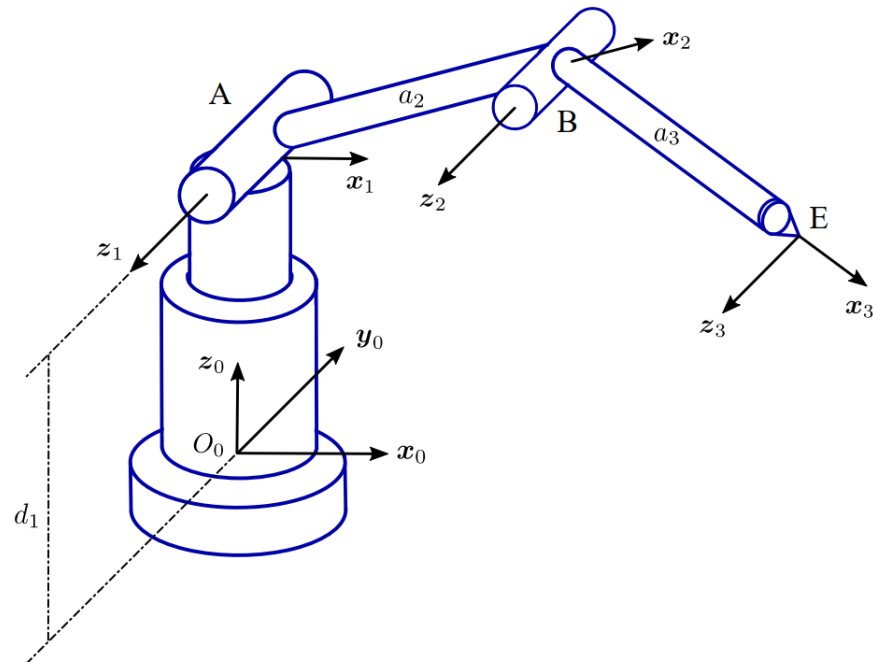
تمرین کلاسی شماره 1

باز می‌شود: یکشنبه، 30 فروردین 1405، 9:00 صبح
مهلت: یکشنبه، 30 فروردین 1405، 6:00 عصر

نمره

خلاصه نمره‌دهی

پنهان از شاگردان	خیر
شرکت کنندگان	40
تحویل داده شده	0
نیازمند نمره‌دهی	0



Examples:

✓ Example 3: SCARA Manipulator

- Algorithmic Approach (Paul's Convention)

Denote $\mathbf{q} = [\theta_1, \theta_2, d_3, \theta_4]^T$ and $\mathbf{x} = [x_E, y_E, z_E, \Theta_E]^T$

Affix the frames and find DH-parameters.

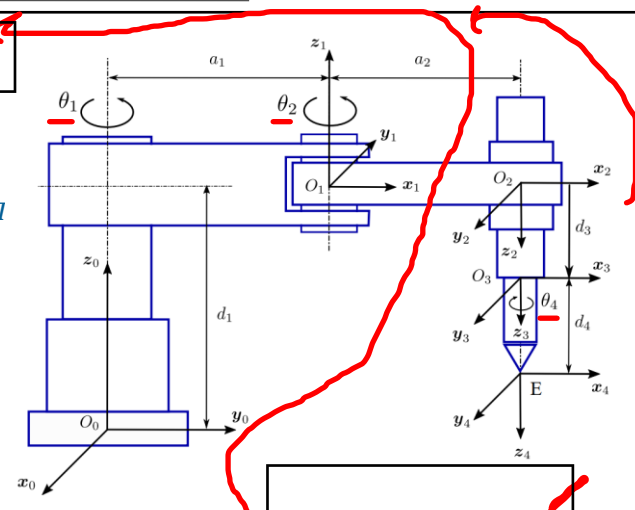
Find the homogeneous transformations:

$${}^0_1T = \begin{bmatrix} c_1 & -s_1 & 0 & a_1c_1 \\ s_1 & c_1 & 0 & a_1s_1 \\ 0 & 0 & 1 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}, {}^1_2T = \begin{bmatrix} c_2 & s_2 & 0 & a_2c_2 \\ s_2 & -c_2 & 0 & a_2s_2 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

$${}^2_3T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}, {}^3_4T = \begin{bmatrix} c_4 & -s_4 & 0 & 0 \\ s_4 & c_4 & 0 & 0 \\ 0 & 0 & 1 & d_4 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Find the loop closure equation in matrix form:

$${}^0_ET = {}^0_3T = {}^0_1T {}^1_2T {}^2_3T {}^3_4T$$



i	α_i	a_i	d_i	θ_i
1	0	a_1	d_1	θ_1
2	π	a_2	0	θ_2
3	0	0	d_3	0
4	0	0	d_4	θ_4

Forward Kinematics

• Motivating Example

✓ Kinematic loop closure

- Assign base coordinate frame $\{0\}$

Denote $\mathbf{q} = [\theta_1, \theta_2, \theta_3]^T$ and $\mathbf{x} = [x_e, y_e, \theta_e]^T$

Denote the link vectors $\vec{a}_i$ and the end-effector vector $\vec{e}$

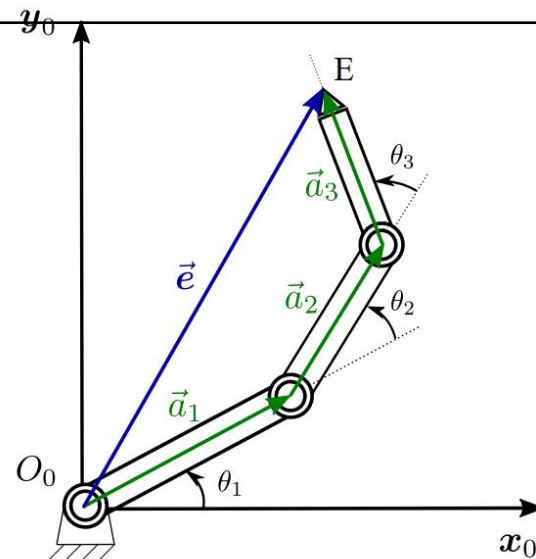
Write the loop closure vector equation:

$$\vec{a}_1 + \vec{a}_2 + \vec{a}_3 = \vec{e}$$

$$a_1 \cos \theta_1 + a_2 \cos(\theta_1 + \theta_2) + a_3 \cos(\theta_1 + \theta_2 + \theta_3) = x_e$$

$$a_1 \sin \theta_1 + a_2 \sin(\theta_1 + \theta_2) + a_3 \sin(\theta_1 + \theta_2 + \theta_3) = y_e$$

$$\theta_e = \theta_1 + \theta_2 + \theta_3$$



Forward Kinematics

- Forward Kinematics: Matrix Loop Closure

- ✓ Key Concepts:

- The 3D Need: Position (X,Y,Z) is not enough; Orientation is critical.
- The Solution: 4×4 Homogeneous Transformation Matrix (HTM).
- Loop Closure: Target Frame = Sequential Product of Joint Matrices.

- ✓ Key Equations:

- 1. End-Effector Target (0T_e):

$${}^0T_e = \left[\begin{array}{ccc|c} & & & {}^0x_e \\ & & & {}^0y_e \\ & & & {}^0z_e \\ \hline 0 & 0 & 0 & 1 \end{array} \right]$$

- Kinematic Chain:

$${}^0T_n = {}^0T_1 \cdot {}^1T_2 \cdots {}^{n-1}T_n$$

- Loop Closure:

$${}^0T_e = {}^0T_n$$

- Examples:

- ✓ Example 4: Planar RRR Manipulator

- Target Frame (Task Space):

- Given workspace vector: $X = [x_e, y_e, 0, 0, 0, \theta_e]$

$${}^0T_e = \left[\begin{array}{ccc|c} c_{\theta_e} & -s_{\theta_e} & 0 & x_e \\ s_{\theta_e} & c_{\theta_e} & 0 & y_e \\ 0 & 0 & 1 & 0 \\ \hline 0 & 0 & 0 & 1 \end{array} \right]$$

- Robot Frame (Joint Space):

- Derived via Denavit-Hartenberg (D-H):

$${}^0T_3 = \left[\begin{array}{ccc|c} c_{123} & -s_{123} & 0 & a_1c_1 + a_2c_{12} + a_3c_{123} \\ s_{123} & c_{123} & 0 & a_1s_1 + a_2s_{12} + a_3s_{123} \\ 0 & 0 & 1 & 0 \\ \hline 0 & 0 & 0 & 1 \end{array} \right]$$

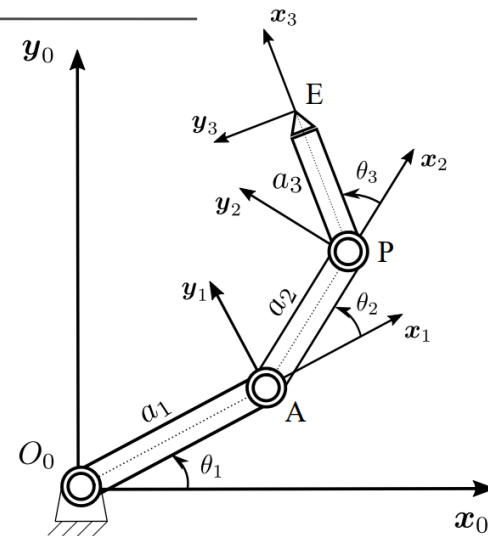
- Loop Closure (${}^0ET = {}^0_3T$):

- Equating corresponding elements yields:

$$x_e = a_1c_1 + a_2c_{12} + a_3c_{123}$$

$$y_e = a_1s_1 + a_2s_{12} + a_3s_{123}$$

$$\theta_e = \theta_1 + \theta_2 + \theta_3$$



ARAS
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Thank You

